

Statistical Inference for Linear Stochastic Approximation with Richardson–Romberg Extrapolation

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Motivation: Confidence Intervals for Dependent Algorithms

Stochastic approximation

Updates arrive sequentially and depend on data generated by a Markov chain.

Constant step size

Stable and fast in practice, but leaves a persistent step-size bias.

Statistical inference

We need confidence intervals, not only point estimates.

The main difficulty is not only convergence:

biased center + dependent noise = unreliable confidence intervals

This thesis studies how Richardson–Romberg extrapolation can remove the leading constant-step bias while preserving the usual long-run covariance target for inference.

Linear Stochastic Approximation

We observe a Markov chain $Z_1, Z_2, \dots$ and run

$$\theta_k = \theta_{k-1} - \alpha(A(Z_k)\theta_{k-1} - b(Z_k)).$$

The target is the equilibrium of the averaged problem:

$$\theta^* = \bar{A}^{-1}\bar{b}, \quad \bar{A} = \int A(z)\pi(dz), \quad \bar{b} = \int b(z)\pi(dz).$$

Why average iterates?

Polyak–Ruppert averaging reduces variance and gives the standard $\sqrt{n}$ scale.

Why constant steps?

They are practical and stable, but the estimator is generally centered at $\theta^* + O(\alpha)$ instead of θ^* .

If the interval center is shifted by the step-size bias, even a good variance estimate cannot restore correct coverage.

Richardson–Romberg in the Step Size

Run two averaged LSA trajectories on the **same Markov chain path**:

$$\bar{\theta}_n^{\text{RR}} = 2\bar{\theta}_n^{(\alpha)} - \bar{\theta}_n^{(2\alpha)}.$$

Single constant step

$$\bar{\theta}_n^{(\alpha)} \approx \theta^* + \alpha\Delta + \text{noise}$$



RR combination

$$2(\alpha\Delta) - (2\alpha\Delta) = 0$$

RR is used to move the center of the confidence interval, not to estimate the covariance. Covariance estimation is a separate problem.

The shared trajectory is important: the two branches have highly correlated noise, so the extrapolation targets bias rather than adding an independent simulation error.

What Distribution Should We Approximate?

For a direction u , the practical statistic is

$$\sqrt{m}u^\top (\bar{\theta}_{n,n_0}^{\text{RR}} - \theta^*), \quad m = n - n_0.$$

The target variance is the standard Markovian long-run covariance:

$$\Sigma_\infty = \bar{A}^{-1} \Sigma_\varepsilon^{\text{M}} \bar{A}^{-T}.$$

Markovian noise

$\Sigma_\varepsilon^{\text{M}}$ includes serial correlations of $\varepsilon(Z_k)$.

Inference goal

Approximate the scalar statistic by $\mathcal{N}(0, u^\top \Sigma_\infty u)$.

Main Theoretical Result

Under uniform geometric ergodicity, Lyapunov stability, bounded noise, and a mixing-scale burn-in, the burned-in RR average satisfies a Berry–Esseen bound.

$$d_K \left(\sqrt{m} u^\top \frac{\bar{\theta}_{n,n_0}^{\text{RR}} - \theta^*}{\sigma(u)}, \mathcal{N}(0, 1) \right) \leq C(u, c, \theta_0) \text{polylog}(n) n^{-\frac{1}{4}}.$$

Balanced scale:

$$\alpha = cn^{-\frac{1}{2}}, \quad n_0 \approx (\alpha a)^{-1} \log^2 n, \quad \sigma^2(u) = u^\top \Sigma_\infty u.$$

The theorem turns the RR averaged estimator into a valid normal-approximation object for scalar confidence intervals.

Proof Idea Without the Technical Details

1. Deterministic RR weights

The leading averaged error is a weighted sum of Markovian noise. RR changes the deterministic kernel, and the kernel is close to $\bar{A}^{-1}$.

2. Markovian noise reduction

A Poisson-equation decomposition converts the main dependent sum into a martingale term plus controlled boundary terms.

3. Variance comparison

The finite-window variance of the RR statistic is compared with Σ_∞ .

4. Remainder control and burn-in

Misadjustment, random initial products, and deterministic transients are shown to be smaller than the leading Gaussian term.

The proof separates what RR cancels from what normal approximation and mixing have to control.

From Theory to Confidence Intervals

In practice Σ_∞ is unknown, so the long-run variance must be estimated from one dependent trajectory.

OBM

Overlapping batch means estimate the long-run variance of the projected trajectory.

OBM-LW / lugsail

A signed two-window correction can reduce the leading OBM window bias.

RR changes the center; OBM and lugsail change the width.

The thesis proves the distributional approximation for the RR estimator. A full non-asymptotic theory for OBM and lugsail along RR trajectories is left as a separate future direction.

Numerical Experiments

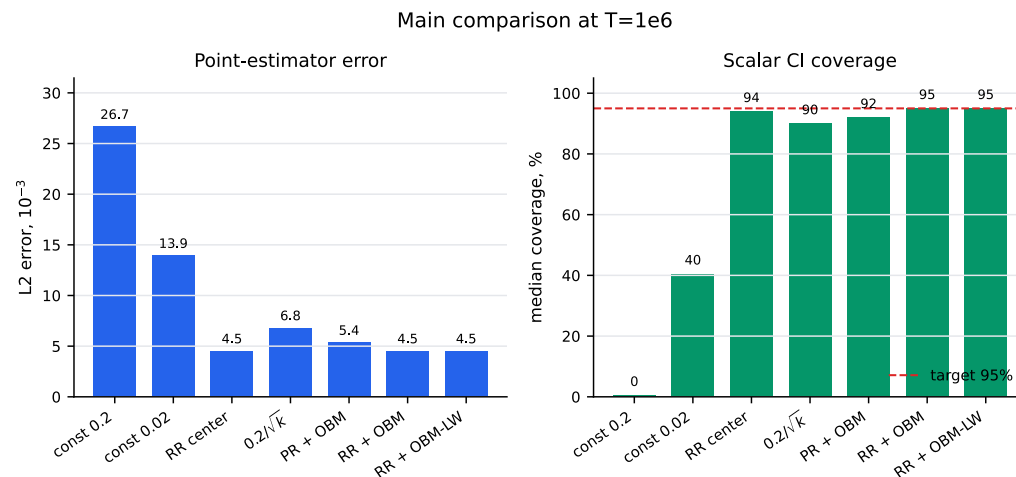
Quantity	Main setup
Problem class	Finite-state Markovian LSA
Dimension and states	$d = 5, 10$ Markov states
Monte Carlo design	100 problems, 100 trajectories per problem
Trajectory length	$T = 10^6$
RR stepsizes	$\alpha \in \{0.2, 0.02\}$ in the main comparison
Inference	95% scalar intervals in a random projection direction
Variance estimators	Batch means, OBM, OBM-LW, MSB, and oracle diagnostics

The experiments are designed to separate the point-estimator bias from the long-run variance estimation error.

Main Comparison: RR Fixes the Center

Method	L2	Cov.
$\alpha = 0.2$ const.	26.67	0.5%
$\alpha = 0.02$ const.	13.93	40.5%
RR const.	4.52	94.0%
Diminishing	6.80	90.0%
PR + OBM	5.35	92.0%
RR + OBM	4.52	95.0%
RR + OBM-LW	4.52	95.0%

L2 is reported in units of 10^{-3} ; medians over 100 problems.



The coverage gain is not achieved by widening intervals: RR mainly improves the interval center by reducing step-size bias.

Diagnostics: Stepsize and Variance Estimation

RR pair	Single-step coverage	RR L2	RR cov.
(0.04, 0.02)	0.04: 92.0%; 0.02: 94.0%	2.97	94.0%
(0.10, 0.05)	0.10: 86.0%; 0.05: 91.5%	2.97	94.0%
(0.20, 0.10)	0.20: 67.5%; 0.10: 86.0%	2.97	94.0%

For the largest adjacent pair, oracle and practical variance intervals are almost identical:

RR + oracle variance: 95.0% coverage; RR + OBM: 95.0%; RR + MSB: 95.0%.

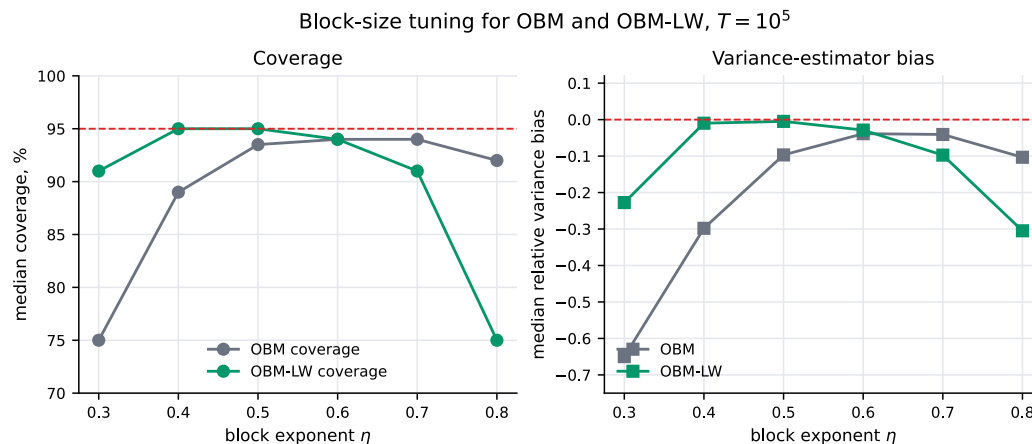
In this regime, long-run variance estimation is not the bottleneck; the key effect is still the RR bias cancellation in the center.

Variance Estimation: When Does Lugsail Help?

OBM can be too narrow when the Bartlett-window bias is negative.

Lugsail can help by cancelling the leading window-bias term, especially for smaller block sizes or more persistent trajectories.

But it is not automatic: large blocks can make the signed correction unstable.



Step-size RR and lugsail OBM solve different problems: center bias versus variance-estimator window bias.

Contributions and Limitations

Main contributions

- Formulated RR-averaged constant-step LSA inference under Markovian noise.
- Proved a scalar Berry–Esseen bound at the balanced scale.
- Transferred the result from stationary starts to burned-in deterministic starts.
- Verified the bias-correction mechanism in finite-state experiments.

Limitations and future work

- Full multivariate confidence regions.
- Non-asymptotic OBM and lugsail theory along RR trajectories.
- Sharper dependence on mixing time and stability constants.
- More diagnostics for slow mixing and matrix covariance estimates.

The central message: RR makes constant-step LSA statistically usable by correcting the center, while standard long-run variance tools handle the interval width.

Thank you!

Questions?